

CHANGSHA HUANGHUA INTL, China

WMO: 592871

Lat: 28.189N

Lon: 113.220E

Elev: 66

StdP: 100.53

Time Zone: 8.00 (E08)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.9	-0.8	-8.0	1.9	7.4	-5.2	2.5	5.9	7.4	5.1	7.0	4.6	2.0	340	0.388

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.4	37.1	26.7	36.1	26.6	35.1	26.4	28.7	34.0	28.0	33.0	27.4	32.2	3.4	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.2	23.2	31.2	26.9	22.8	30.9	26.1	21.7	30.1	93.2	34.3	90.1	33.1	87.5	32.4	30.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
7.6	6.6	5.8	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-4.0	39.1	1.4	1.1	-5.0	39.9	-5.8	40.5	-6.5	41.1	-7.5	41.9		
			-4.4	29.1	1.4	1.1	-5.4	29.9	-6.2	30.5	-7.0	31.1	-8.0	31.9		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	18.6	5.5	7.9	13.0	18.9	23.3	26.7	30.0	29.4	25.5	20.0	13.9	7.9
	DBStd	9.06	3.53	4.66	4.56	4.11	3.09	2.81	2.40	2.76	3.47	3.23	3.99	3.04
	HDD10.0	343	146	88	21	1	0	0	0	0	0	0	11	77
	HDD18.3	1392	398	292	174	42	2	0	0	0	1	19	141	322
	CDD10.0	3468	6	31	114	267	412	502	621	600	463	311	128	13
	CDD18.3	1476	0	2	9	58	156	252	363	342	215	72	7	0
	CDH23.3	15568	1	16	86	446	1205	2474	4740	4124	1954	467	54	0
	CDH26.7	7283	0	3	19	133	454	1059	2534	2092	849	132	8	0
Wind	WSAvg	2.5	2.5	2.6	2.4	2.5	2.4	2.2	2.7	2.6	2.5	2.5	2.4	2.5
Precipitation	PrecAvg	1508	77	90	152	185	214	231	150	122	82	64	86	55
	PrecMax	1813	181	194	369	275	453	492	392	337	244	156	205	140
	PrecMin	1017	16	13	91	82	57	112	11	28	2	3	12	0
	PrecStd	225	40	41	61	58	86	91	99	67	55	36	60	36
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	20.1	25.9	29.1	32.1	34.2	36.1	38.2	38.8	36.9	32.9	27.8	20.8
		MCWB	12.5	17.4	20.8	22.3	24.5	26.3	26.1	25.9	26.8	22.8	18.8	13.0
	2%	DB	16.0	21.8	25.8	30.0	33.0	35.0	37.2	37.1	35.0	29.9	24.9	17.9
		MCWB	9.4	15.0	18.2	21.4	23.5	26.2	26.5	26.9	25.8	20.9	17.4	11.1
	5%	DB	13.2	18.1	23.0	28.1	31.1	33.9	36.2	36.1	33.2	27.9	22.2	15.1
		MCWB	8.6	13.0	16.5	20.2	23.1	26.0	26.6	26.6	25.1	19.7	15.9	9.7
	10%	DB	11.0	14.9	20.1	26.0	29.2	32.1	35.2	34.9	31.2	26.0	20.1	13.0
		MCWB	7.6	10.9	15.2	19.6	22.6	25.4	26.6	26.6	23.9	18.9	15.4	8.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.3	18.7	23.0	24.9	27.0	28.7	29.6	29.5	28.7	24.8	21.3	14.8
		MCDB	19.2	24.5	27.0	30.6	31.8	33.7	34.8	34.8	33.7	28.2	25.2	19.0
	2%	WB	11.0	15.9	19.6	23.0	25.6	27.8	28.7	28.7	27.4	23.0	19.2	12.4
		MCDB	14.2	20.7	24.4	27.5	30.3	32.2	33.9	34.1	32.1	26.8	21.8	16.3
	5%	WB	9.4	13.6	17.7	21.7	24.6	26.9	28.0	28.0	26.4	21.6	17.7	10.8
		MCDB	12.2	16.6	21.1	25.8	28.8	31.1	32.9	33.2	30.8	25.1	20.3	13.4
	10%	WB	7.7	11.7	16.0	20.6	23.7	26.2	27.4	27.4	25.6	20.5	16.4	9.7
		MCDB	9.9	14.5	19.2	24.3	27.5	30.3	32.2	32.1	29.6	24.0	19.2	11.8
Mean Daily Temperature Range	5% DB	MDBR	6.3	6.8	7.8	8.7	8.3	7.4	8.4	8.6	8.5	8.7	8.4	7.9
		MCDBR	13.1	13.7	13.8	12.9	12.2	10.1	10.2	10.7	11.3	12.6	13.2	13.8
		MCWBR	7.2	7.4	6.5	5.7	4.9	3.4	2.9	3.1	3.9	4.7	6.2	7.4
	5% WB	MCDBR	10.5	11.4	11.0	10.5	9.7	8.8	9.1	9.7	9.8	9.3	9.0	11.1
		MCWBR	6.3	6.6	6.1	5.5	4.8	3.8	3.5	3.7	4.1	4.5	5.0	6.4
Clear-Sky Solar Irradiance	taub	0.695	0.736	0.787	0.768	0.696	0.678	0.607	0.673	0.699	0.735	0.682	0.664	
	taud	1.495	1.459	1.408	1.481	1.646	1.698	1.874	1.698	1.622	1.500	1.566	1.563	
	Ebn at Noon	540	567	577	612	662	671	719	667	628	559	546	538	
	Edh at Noon	255	285	316	301	256	242	202	240	251	269	235	229	
All-Sky Solar Radiation	RadAvg	1.71	2.04	2.70	3.36	3.86	4.11	5.29	4.57	3.84	3.03	2.38	1.98	
	RadStd	0.38	0.58	0.30	0.43	0.56	0.49	0.52	0.39	0.41	0.47	0.42	0.44	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (6 neighbors)	+0.29	N/A	N/A	+0.43	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page